Finite subgroup atlases: proofs, certified tables, and source qualifications

L054 synthesis of acquired components

This manuscript concerns Connor and Leemans' seven-page author edition of *An atlas of subgroup lattices of finite almost simple groups* (arXiv version 2, 29 November 2013; PDF dated 2 December 2013), and collates its eight-page 2015 journal version, pages 259–266. The maximal descent and recursive naming constructions are the authors' ideas. The arguments below and the acquired complete proofs make their interfaces explicit and correct several ancillary formulations. No novelty or new historical measurement is asserted.

Main results and the atlas boundary. We recover the maximal-descent and structural-certificate constructions under explicit complete local subgroup and exact-operation interfaces. The acquired classified roster contains all 127 almost simple isomorphism classes of order at most one million. Complete numerical tables and structural certificates have been checked for every one of these groups, including 47 groups absent from the current small companion roster. Seven larger applications are also complete. Their sizes are recorded below.

Domain	Groups	Subgroup-class rows
All almost simple groups of order at most 10^6	127	10,694
Seven specified larger groups	7	5,294
Complete reconstruction	134	15,988

These are row occurrences in actual tables, not pairwise different abstract subgroups. Of these rows, 13,520 belong to 87 current companion tables; one printed label in the A_{11} table is refuted and replaced by its verified structure. The remaining 2,468 rows are additional complete applications with generated structural descriptions.

This proves the small-order claim for a domain containing the source's 2012 online-ATLAS domain and supplies more than one hundred complete tables. It is *not* a reproduction or verification of the entire named historical or current atlas. The original articles give no exact filename roster. The current inventory of 116 companion tables, specifically attributed outputs such as the O'Nan table, and unlinked numerical claims remain in a separate ledger. No unpaid current table acquires its structural labels through this main-result reconstruction. Historical file identity and the reported historical timings are not asserted.

1 The maximal-descent and weight argument

All groups here are finite. For $H \leq G$, write $[H]_G$ for its conjugacy class, ordered by containment of representatives. Suppose each processed H comes with an actual isomorphism $\phi_H : H \to \widetilde{H}$, a complete cover of its maximal-subgroup conjugacy classes, and exact ambient conjugacy tests. Lists may repeat a local class; duplicates must first be identified by actual local conjugacy. Completeness is an explicit mathematical/software input. It does not follow from a plausible list length or matching orders.

Theorem 1. *Maintain a permanent registry of both processed and pending classes, initially containing G. Process each registered parent once; pull back its complete local maximal cover, fuse the resulting subgroups under conjugacy in G, and register each unseen class. This terminates with exactly all subgroup classes of G. Reachability among its actual maximal-inclusion edges is the entire class containment relation.*

Proof. An isomorphism and its inverse preserve subgroup containment, maximality, conjugacy and normalizers. Hence every new class and edge is genuine. There are finitely many classes, and no registered class is processed twice. For completeness, refine any $K \leq H$ to a maximal subgroup chain. At each step a member of the complete local cover is conjugate inside the parent to the next chain term. Its ambient fusion only changes the representative by conjugacy. Induction down this chain, starting with G, discovers $[K]_G$. The same argument inside any larger representative proves that containment implies reachability. The reverse implication follows from transitivity of containment on classes. □

The permanent registry repairs the source prose's test against only the current queue unless an additional scheduling invariant is supplied. Already completed parents can be reused: their complete transported local lists and exact fusion remain valid, while every unprocessed parent is still an obligation. The acquired literal-state and continuation interfaces implement this invariant. Named display strings alone are not embedded subgroup data; a failed symmetric-group checkpoint demonstrated why actual generators must be serialized.

For representatives H, K, let $M_1, \dots, M_t$ be the distinct local H-classes of maximal subgroups that fuse to $[K]_G$. Put

$$c_H = [G : \mathrm{N}_G(H)], \qquad a_{HK} = \sum_{j=1}^{t} [H : \mathrm{N}_H(M_j)], \qquad b_{KH} = c_H a_{HK}/c_K.$$

Orbit–stabilizer proves that a_{HK} counts the actual maximal embeddings inside H. Counting the pairs of conjugates (H', K') with K' maximal in H' in both orders proves the expression for b_{KH} and its integrality. Local classes that fuse must be added, rather than counted once. Duplicate local representatives must not be added twice. The exact bounds $a_{HK} \leq c_K$ and $b_{KH} \leq c_H$ follow immediately.

The class-poset covers are obtained by transitive reduction of the complete maximal-edge graph. On a cover, every embedding is maximal, so these weights are exactly the source's n_{HK}, n_{KH}. Outside covers, the displayed maximal-edge weights do not automatically count every embedding. The acquired robust local-list proof and retained-inventory reweighting pay these distinctions. In particular, an early duplicate local-class output for the $\mathrm{P\Gamma U}_3(5)$ model inflated one computed weight from $(3, 1)$ to $(6, 2)$; the corrected complete replay matches the source. This was our execution issue, not a source-table error.

2 Recursive names as exact existence assertions

A structural expression is interpreted recursively. Parentheses in the native source are retained; otherwise products associate to the left. A cross denotes a direct product, a colon a split extension, and the centered dot a nonsplit extension. The source also uses a different full stop for an extension whose splitting has not yet been specified. Names without the action or factor data need not determine a unique group.

For $N \lhd H$, $Q = H/N$, choose a section $s : Q \to H$ with $s(1) = 1$. Set $\alpha_q(n) = s(q)ns(q)^{-1}$ and $f(q, r) = s(q)s(r)s(qr)^{-1} \in N$. Then $N \times Q \to H$, $(n, q) \mapsto ns(q)$, is bijective, and multiplication is

$$(n, q)(m, r) = (n\alpha_q(m)f(q, r), qr).$$

The acquired complete extension proof verifies associativity exactly from $\alpha_q\alpha_r = \mathrm{Inn}(f(q,r))\alpha_{qr}$ and $f(q,r)f(qr,t) = \alpha_q(f(r,t))f(q,rt)$, with the identity normalizations. An explicit complement proves splitting; a commuting complement proves a direct product. Failure of a search to find a complement proves neither nonsplitting nor absence of a specified structure.

At each nonterminal naming step choose a proper nontrivial normal subgroup. Both child orders are smaller, so recursion terminates. Exact leaf identifications and the verified extension data reconstruct the parent by induction. Without those data the expression records only an existence assertion. The source's preferences for short or recognizable factors may select one such description, but are not a canonical-isomorphism theorem.

Every accepted row uses an actual embedded subgroup. An earlier named model is transported by an explicit bijection fixing identity and respecting right multiplication by every ordered generator. Induction on words proves that this is a homomorphism; bijectivity proves it is an isomorphism. Complete Cayley-word maps, faithful smaller actions, elementary normal forms, actual normal factors and complements, and exhaustive quotient-lift obstructions supply the concrete certificates. Each criterion has its acquired complete proof. Source typography is checked separately: a flattened superscript cannot silently become a cyclic order, and a dot cannot silently be replaced by a colon. Repeated descriptions may use different concrete models, each with its own exact transport.

3 Finite application and retained inputs

The attached current small-table roster has 80 tables and 8,226 subgroup-class rows. Each numerical route fixes an actual named group, a complete subgroup inventory, actual embedded generators, local class fusion, both incidence weights, and a bijection to the native source rows. Each structural route checks the source expression and every required normal factor, quotient, complement or nonsplitting obstruction. Their complete proof texts and separate author and reviewer records are acquired without another census.

Theorem 2. *Subject to the acquired classification of finite simple groups, their exact orders and outer automorphism groups, there are 127 almost simple isomorphism classes of order at most one million. For each class, the acquired applications give its full subgroup-class inventory, maximal incidences and a certified structural description of every subgroup class. Together these tables contain 10,694 subgroup-class rows.*

Proof. The acquired small-order roster proof first bounds the simple socle using the exact family orders and the retained classified exceptional list. There are 56 socles in this range. For a fixed socle S, an almost simple group is the full inverse image of a subgroup of $\mathrm{Out}(S) = \mathrm{Aut}(S)/S$. An isomorphism preserves the characteristic simple socle and therefore identifies such groups exactly by the corresponding outer-subgroup conjugacy relation. The complete small outer-group calculations give 127 cases. The complete roster proof, classification locators and arithmetic checks are included in the proof collection; no new proof of the classification is claimed here.

The 80 current small-table applications cover their exact named targets. The remaining 47 cases have 28 complete parent descents and 19 exact index-two restrictions from those parent inventories. The descent theorem above and the acquired restriction theorem prove completeness. Explicit named-model identities join each group to its roster position. Complete generator-edge transports, normal factors, complements and nonsplitting obstructions prove the 2,468 additional structural rows. Thus all 127 roster positions have complete numerical and structural applications. □

Proposition 3. *The following seven larger groups have complete numerical tables and structural certificates, with the stated correction for A_{11}.*

Group	Subgroup-class rows
A_{10}	430
A_{11}	788
M_{23}	204
$\mathrm{PSp}(4,5)$	307
$\mathrm{PSp}(4,7)$	603
$\mathrm{Sp}(6,2)$	1,369
S_{10}	1,593

Proof. Each application acquires a complete numerical inventory and an exact bijection to the companion rows, then verifies every displayed structural assertion with the literal subgroup generators. The acquired mathematical proofs justify the reused transport, support, extension and finite normal-form tests. The seven independent table reviews bind all 5,294 rows and their native expressions. The A_{11} review recovers 787 printed labels and proves the remaining printed label false, supplying its actual structure instead. The exact correction is stated below. No inference from an unverified group name or from matching orders substitutes for these certificates. □

Only M_{23} is a sporadic group among these seven larger applications. We do not infer coverage of the source's plural historical larger-sporadic examples from that one case. The quantified main-result reconstruction and the correctness of every individually named historical output are separate claims. The latter is not made here.

The source's Section 4 reports all groups in its ATLAS-restricted small-order domain, some larger groups, and more than one hundred complete outputs. The 127-group theorem proves a stronger small-order mathematical statement; the seven further applications supply explicit larger cases. No historical membership assumption is needed for that quantified conclusion. Conversely, these conclusions do not validate every current companion PDF. The retained index has 133 group rows, 130 linked occurrences and 116 distinct primary links, with a printed update date of 4 August 2013. Its documentary date does not establish historical byte identity. The O'Nan landing page specifically attributes its current linked table to this article's programs; that particular companion obligation remains recorded and unpaid. The source-claim map preserves all such boundaries. The companion index bounds the simple socle order $|S|$, whereas the article bounds $|G|$; our small-order theorem does not establish that stronger companion domain. The unlinked Suzuki and Fischer numerical assertions are retained separately. The completed S_{11} numerical inventory remains additional evidence with its structural labels unproved.

Named constructors, complete external local covers or previously accepted ambient inventories, and the exact group-operation interfaces remain explicit premises. Source counts are not used as completeness filters. All earlier mathematical authorship and distinct reviews are retained; this synthesis neither repeats the prior proofs nor represents itself as an independent review of its own earlier components.

4 Exact source corrections and attribution

Class posets and joins. The subgroup join is $\langle H, K\rangle$, not ordinary set union. The conjugacy-class poset need not be a lattice, contrary to the literal sentence on author page 2 and journal page 261. For an elementary counterexample take $G = A_5$. The classes of subgroups C_2 and C_3 have both S_3 and A_4 as upper bounds: embed S_3 on three letters and compensate every odd permutation by swapping the other two; embed A_4 on four letters. The relevant C_2 subgroups are conjugate in A_5, as are the C_3 subgroups. Indeed an odd conjugator may be made even using an odd permutation centralizing the relevant element. Any least upper bound would lie below both S_3 and A_4, and its order would be divisible by six and divide six. It would therefore

have order six. But A_4 has no subgroup of order six: such a subgroup would have index two, whereas every homomorphism $A_4 \to C_2$ kills the generating 3-cycles. Thus no such least upper bound exists. The accepted literal A_5 catalogue provides the same incidence obstruction.

Composition and deterministic names. Strict normal inclusions alone do not define a composition series: the factors must be simple, equivalently the lower term is maximal normal in the next one. Jordan–Hölder gives the multiset of simple factors up to isomorphism, not uniqueness of the intermediate subgroups. For example, C_2^2 has three different order-two intermediate terms with the same two composition factors. These are qualifications to author page 4 and journal page 263, not new classification results.

The impossibility sentence about deterministic names on those pages is also too strong literally. Number the identity by zero, enumerate all remaining labelings of a finite group's complete multiplication table, and take the lexicographically least resulting table as its name. Isomorphic groups yield the same collection of relabeled tables; equality of the minima supplies an isomorphism. This finite deterministic construction is not an efficient or useful short notation. It shows that ambiguity of factor descriptions is not undecidability of finite group identification. The source's example of nonisomorphic groups with the same normal-section information remains attributed to its earlier cited paper.

The group-name chain. Both native author page 3 and journal page 262 print a chain beginning $S_4(3) \cong U_4(2)$ and including $O_6^-(3)$ and $W(E_6)$. As printed the chain is false. The independently acquired named $\mathrm{PSU}(4,2)$ certificate has order 25920, whereas the acquired E_6 Weyl calculation has order $27 \cdot 1920 = 51840$. These orders preclude the asserted isomorphism with $W(E_6)$. We do not silently replace the field parameter or a full Weyl group by an index-two subgroup. The valid general point—that one group may have several geometric descriptions—does not require that faulty chain.

The A_{11} companion correction. Row 186 prints the left-associated expression $(2.A_5) : 2$. Its order-120 normal section would be the nonsplit double cover of A_5. The independently reviewed application proves that this section cannot embed in A_{11} under the stated convention, and checks the actual order-240 subgroup by a complete literal certificate. Its structure is $(A_5 \times C_2) \cdot C_2$, equivalently $C_2 \cdot (A_5 : C_2)$. The printed expression is retained as a refuted source assertion; it is not silently reassociated. This correction concerns a companion label, not a newly solved open conjecture.

Source examples and historical reports. The examples $S_5 = A_5 : C_2$ and $A_4 = C_2^2 : C_3$ have their standard actual normal factors and complements, also present in the acquired registry. The groups S_5 and $A_5 \times C_2$ have the same two composition factors but centers of orders one and two, respectively, so are not isomorphic. A coordinate flag of C_2^5 gives the displayed product of five order-two factors, each split and commuting. The label $5 \times 2 \cdot 2 : 2$ can describe both D_{40} and $C_5 \times D_8$: use their cyclic order-20 normal subgroup, its order-10 subgroup, and an order-two complement. The intermediate index-two extension is nonsplit. The final action in the first case inverts the order-five factor and in the second case centralizes it. The centers have orders two and ten, respectively, so the groups are not isomorphic. This pays the source's concrete illustration of nonunique notation.

The introduction's earlier sporadic subgroup computations, the small-group count near order 1024, software-database descriptions and historical normal-section example remain attributed antecedents. The concrete $L_3(7) : 2$ degree-reduction and machine-time/memory report, and the journal's qualitative performance comparison, are empirical source reports. Correctness of the algorithm does not prove a universal speed or memory advantage, and no historical timing is reproduced or credited here. The source-clause register lists every such boundary separately.

Exact current small-table roster

Current file locator	Group order	Rows
alt5	60	9
sym5	120	19
psl27	168	15
pgl27	336	23
alt6	360	22
psl28	504	12
psl211	660	16
m10	720	25
pgl29	720	26
psigmal29	720	56
psl213	1092	16
pgl211	1320	29
pgammal29	1440	65
psigmal28	1512	25
pgl213	2184	30
psl217	2448	22
alt7	2520	40
psl219	3420	19
psl216	4080	21
pgl217	4896	32
sym7	5040	96
psl33	5616	51
psu33	6048	36
psl223	6072	23
pgl219	6840	36
psl225	7800	37
m11	7920	39
psl216d2	8160	47
psl227	9828	16
psl33d2	11232	78
pgammau33	12096	100
pgl223	12144	34
psl229	12180	22
psl231	14880	29
pgl225	15600	43
psigmal225	15600	103
psl225d2	15600	39
pgammal216	16320	69
pgl227	19656	32
alt8	20160	137
psl34	20160	95
psu42	25920	116
suz8	29120	22
psigmal227	29484	38
pgl231	29760	44
pgammal225	31200	115
psl232	32736	24
psigmal34	40320	152
psl34d21	40320	151
psl34d23	40320	102
sym8	40320	296
pgammau42	51840	350
pgammal227	58968	67
pgl34	60480	100

Current file locator	Group order	Rows
psu34	62400	34
psl34d2e2	80640	288
suz8d3	87360	39
m12	95040	147
pgammal34	120960	226
psl34d6	120960	143
psl34ds3	120960	142
psu34d2	124800	80
psu35	126000	80
pgammal232	163680	30
j1	175560	40
alt9	181440	223
m12d2	190080	213
psl34dD12	241920	409
psigmau34	249600	120
psigmau35	252000	148
psl264	262080	76
sym9	362880	554
psl35	372000	140
pgu35	378000	93
m22	443520	156
j2	604800	146
psl35d2	744000	208
pgammau35	756000	244
m22d2	887040	490
sp44	979200	496
Total: 80 tables		8226